Depropanizer Column Ver 1 · OTS

DC3-V1-EE-LL-001

Electrical load list

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
Document number	DC3-V1-EE-LL-001
Title	Electrical load list
Discipline	Electrical
Originator / DRAWN	Apinan Soottitantawat
Date	25-OCT-26
Source	pump hydraulics of the simulator model (DC3-V1-PE-HMB-001) and the assumptions listed; tabs/eelec/work/loads.json

Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
0	25-OCT-26	Issued for teaching	A. Soottitantawat		

Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

Contents

1 · Purpose, scope and basis 3
2 · Load list 4
3 · Load summary and maximum demand 5
4 · Motor feeder cable sizing 6
5 · Hazardous-area classification 7
6 · What fails on power loss 9
7 · Checks against the model and the other documents 9
8 · Assumptions 10
9 · Engineering queries raised 10

1 · Purpose, scope and basis

This document is the electrical load list of the C-201 depropanizer unit: every consumer of electric power with its rated and absorbed power, efficiency, power factor, voltage, bus and duty, the load summary with the maximum demand, the sizing of the motor feeder cables, the hazardous-area classification with gas group, temperature class and equipment protection, and what fails on a loss of power. The pump powers come from the pump hydraulics of the simulator model (DC3-V1-PE-HMB-001); everything else is an assumed allowance, stated as such.

Basis

- 400 V 3-phase 50 Hz for motors up to 200 kW; 230 V for UPS-fed systems and lighting (basis of design).
- Motors IE3 2-pole: 75 kW η 0.95 / PF 0.88; 30 kW η 0.933 / PF 0.87 (assumed, constant at part load).
- Pump efficiency 0.65 (P-201) and 0.6 (P-202) (assumed); motor sized at the rated point (110 % of the maximum-feed flow) with the API 610 margin.
- Demand factors: continuous 1.0, intermittent 0.3, standby 0.1 (assumed). UPS efficiency 0.90, 30 min battery (assumed).
- P-201B is driven by a steam turbine: no electric load (diverse drive); its trip-and-throttle solenoid and governor are on UPS-DCS.

2 · Load list

Tag	Service	Type	V	Bus	Duty	Rated	Absorbed at design kW	Max feed kW	Rated point kW	Load factor	η	PF	Input normal kW	Input peak kW	kVA normal	Source / assumption
P-201A	Feed pump	Induction motor, LV, DOL	400	A	continuous	75 kW	49.7	54.1	56.2	66 %	0.950	0.88	52.4	57.0	59.5	pump hydraulics DC3-V1-PE-HMB-001 (Q × ΔP ÷ η 0.65 assumed); motor η and PF IE3 2-pole (assumed)
P-202A	Reflux / product pump	Induction motor, LV, DOL	400	A	continuous	30 kW	20.7	21.7	22.0	69 %	0.933	0.87	22.2	23.3	25.5	pump hydraulics DC3-V1-PE-HMB-001 (Q × ΔP ÷ η 0.6 assumed); motor η and PF IE3 2-pole (assumed)
P-202B	Reflux / product pump (spare)	Induction motor, LV, DOL	400	B	standby	30 kW	20.7	21.7	22.0	69 %	0.933	0.87	22.2	23.3	25.5	pump hydraulics DC3-V1-PE-HMB-001 (Q × ΔP ÷ η 0.6 assumed); motor η and PF IE3 2-pole (assumed)
P-201B	Feed pump (spare), steam-turbine drive	Steam turbine — non-electric (diverse drive)	—	—	standby	—	49.7	54.1	56.2	—	—	—	0.0	0.0	0.0	same pump duty as P-201A; no electric load. Trip-and-throttle solenoid and governor on UPS-DCS
UPS-DCS	DCS controllers, I/O, operator console share, P-201B turbine solenoid and governor	UPS 230 V AC, 30 min battery	230	A+B	continuous	5 kVA	3.0	—	—	—	0.900	0.95	3.3	3.3	3.5	assumed (load 3.0 kW; input = load ÷ 0.90 UPS efficiency)
UPS-SIS	SIS logic solver, SIS I/O, solenoids of XV-201 / XV-202 / XV-209	UPS 230 V AC → 24 V DC, 30 min battery	230	A+B	continuous	2 kVA	1.0	—	—	—	0.900	0.95	1.1	1.1	1.2	assumed (load 1.0 kW)
AH-219/220	Analyzer house: AI-219 / AI-220 gas chromatographs, sample conditioning, HVAC and purge	Analyzer house feeder	400	B	continuous	10 kW	6.0	—	—	—	1.000	0.90	6.0	6.0	6.7	assumed
LTG-200	Area and platform lighting (LED, 10 W/m², assumed; 10 % of fittings with own battery)	Lighting distribution board	230	B	continuous	6 kW	6.0	—	—	—	1.000	0.95	6.0	6.0	6.3	assumed allowance (night load)
SP-200	Small power: 2 × 63 A welding / maintenance outlets, convenience sockets	Small-power board	400	B	intermittent	20 kW	20.0	—	—	—	1.000	0.85	20.0	20.0	23.5	assumed allowance
SH-200	Motor anti-condensation heaters (3 × 0.1 kW, on while the motor is stopped)	Space heaters	230	A	intermittent	0.3 kW	0.3	—	—	—	1.000	1.00	0.3	0.3	0.3	assumed
HT-200	Heat tracing	none	—	—	—	0.0 kW	0.0	—	—	—	—	—	0.0	0.0	0.0	not required: no water-bearing dead legs; the −1 °C feed is dry LPG (assumed)

Changes to the first draft (epc/epc.json): P-202B: bus B (spare on a different bus section from P-202A (recommended; EE-01)); SP-200: added (small-power and welding outlets were missing from the draft (allowance, intermittent)); SH-200: duty → intermittent (the heaters are on only while a motor is stopped); UPS-DCS / UPS-SIS: efficiency and battery added (UPS input = load ÷ 0.90; 30 min autonomy (assumed)); P-201A, P-202A/B: η and PF added (IE3 2-pole motor data (assumed) to turn shaft power into electrical input and kVA).

3 · Load summary and maximum demand

Duty	Consumers	Connected kW / kVA	Input at design kW	Input at maximum feed kW	Demand factor	Maximum demand kW
continuous	6	128.0	91.0	96.7	1.00	91.0
intermittent	2	20.3	20.3	20.3	0.30	6.1
standby	1	30.0	22.2	23.3	0.10	2.2
Total	9	**178.3**	**133.5**	**140.3**		**99.3**

Maximum demand **99.3 kW**, 52.1 kvar → **112.2 kVA** (PF 0.89); with 20 % growth 134.6 kVA (a dedicated transformer would be 160 kVA; the unit takes a share of the plant substation). Running load of the continuous consumers at design: 91.0 kW, of which the two pump motors are 82 %.

UPS

UPS	Load kW	Rating kVA	Load / rating	Battery energy 30 min kWh	Fed loads
UPS-DCS	3.0	5	63 %	1.67	DCS controllers, I/O, operator console share, P-201B turbine solenoid and governor
UPS-SIS	1.0	2	53 %	0.56	SIS logic solver, SIS I/O, solenoids of XV-201 / XV-202 / XV-209

4 · Motor feeder cable sizing

Cu conductors, XLPE 90 °C, 3-core on perforated tray (IEC 60364-5-52 method E, 30 °C table); derating ambient 40 °C × 0.91 and 4 cables grouped × 0.79; $R_{90} = 1.275\ R_{20}$ (IEC 60228), X 0.08 Ω/km (LV) / 0.10 Ω/km (MV); $\Delta V = \sqrt{3} \cdot I \cdot L \cdot (R \cos\varphi + X \sin\varphi)/V$. Limits: running ≤ 5 %, direct-on-line start (6.5 × I_n at PF 0.3) ≤ 15 % (assumed). MV minimum 25 mm² (short-circuit withstand, assumed). Cable lengths assumed (no plot plan, EE-05).

Motor	Rated kW	Length m	I_n 400 V A	LV cable	I_z derated A	ΔV run / start %	I_n 6.6 kV A	MV cable	ΔV run / start %	Chosen
P-201A	75	150	129.5	50 mm²	138	3.97 / 12.26	7.85	25 mm²	0.027 / 0.075	400 V
P-202A	30	160	53.3	16 mm²	72	4.86 / 12.40	3.23	25 mm²	0.012 / 0.033	400 V
P-202B	30	170	53.3	25 mm²	91	3.32 / 9.05	3.23	25 mm²	0.012 / 0.035	400 V

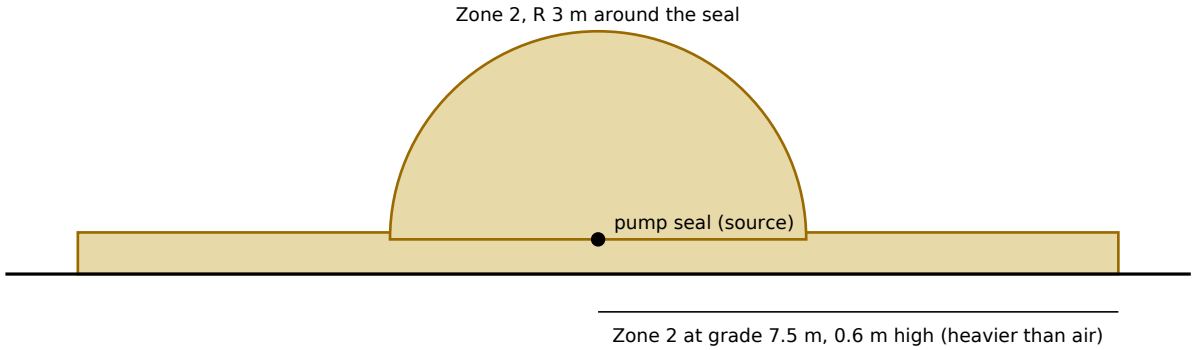
Worked example — P-201A

- Absorbed 49.7 kW (Q·ΔP ÷ η from the model) → rated point 56.2 kW × 1.1 = 61.8 kW → 75 kW motor; input 52.4 kW, 59.5 kVA.
- I_n = 75 000 / (√3·400·0.95·0.88) = 129.5 A; needed table ampacity 180 A → 50 mm² (derated 138 A).
- ΔV over 150 m: running 3.97 %, starting 12.26 % — both inside the limits.
- At 6.6 kV the current is 7.85 A, but the cable is set by the MV minimum and MV switchgear and motor cost far more; MV pays from ≈ 200 kW. 400 V chosen.

5 · Hazardous-area classification

IEC 60079-10-1 approach: source of release → grade (continuous / primary / secondary) → with the ventilation, zone; extents below are typical assumed values for heavier-than-air LPG in open, well-ventilated plant (the release-rate calculation of the standard is not made).

Source of release	Item / location	Grade	Ventilation	Zone	Extent (typical, assumed)	Note
Pump mechanical seals	P-201A/B (V-201 suction 0.42 MPa g, discharge 2.36 MPa g, −1 °C LPG)	secondary (seal failure)	open air, good, availability fair	Zone 2	3 m around the seal; 7.5 m at grade up to 0.6 m (heavier than air)	dual seal with leak detection to flare reduces the release rate
Pump mechanical seals	P-202A/B (1.68 MPa g suction, 39 °C propane)	secondary	open air	Zone 2	3 m around the seal; 7.5 m at grade up to 0.6 m	liquid at its bubble point: flashes on release
Flanges, valve stems, instrument connections	all hydrocarbon piping, FV-201, FV-204, LV-202, LV-203, PV-210	secondary	open air	Zone 2	1 m around the flange / stem; 3 m at grade up to 0.6 m for liquid LPG lines	class 150 / 300 gasketed joints (spiral-wound)
Vessel nozzles, manways, gauge glass	C-201, V-201, V-202; LG-201	secondary	open air	Zone 2	3 m around the vessel envelope; 7.5 m at grade up to 0.6 m	LG-201 glass breakage: armoured gauge with excess-flow ball checks
Relief valve PSV-201	C-201 overhead, discharge to the closed flare header	secondary (body / flanges); no atmospheric discharge	open air	Zone 2	1 m around the valve body; the flare tip is classified by the flare system	closed relief: the outlet is not a source in the unit
Vent to fuel gas	PV-210 / HV-210 on 1"-P-206 to the fuel-gas header (0.40 MPa g)	secondary	open air	Zone 2	1 m around the valve and flanges	closed system; gas group depends on the non-condensables (EE-03)
Sample points	AI-219 / AI-220 sample take-offs, manual LPG sample points (closed-loop cylinder)	primary (each sampling)	open air	Zone 1 + Zone 2	Zone 1 within 1 m, Zone 2 to 3 m; 3 m at grade	closed-loop sampling; sample return to flare / closed drain
Analyzer house	inside AH-219/220 (GC sample lines, fittings)	secondary	artificial ventilation ≥ 6 air changes/h, gas detection	Zone 2 (inside)	the whole house; 1 m around door and HVAC outlets	purge / pressurisation Ex p or Ex ec equipment inside
Pump casing vents and drains	P-201A/B priming vents to flare, low-point drains to the closed drain	secondary (operator action)	open air	Zone 2	1 m around the vent valve; 3 m at grade	priming vents routed to flare, not to atmosphere
Condenser E-202 cooling water	tube leak carries propane into the cooling-water return (10"-CW-402)	secondary, remote	cooling tower	—	classified at the cooling tower (outside this unit)	hydrocarbon detector on the CW return recommended (EM-03)



Gas group and temperature class

Fluid	AIT °C	Gas group	In the unit
propane	450	IIA	yes
n-butane	372	IIA	yes
isobutane	460	IIA	yes
methane (non-condensables)	600	IIA	yes
hydrogen (if present in the non-condensables)	560	IIC	only if confirmed (EE-03)

Lowest AIT 372 °C (n-butane) → hottest allowed temperature class T2 (300 °C); specified **IIA T3** (200 °C) for margin. Hot surfaces: LP steam 175 °C design, C-201 bottoms 109 °C, all below the AIT.

Equipment protection

Equipment	Zone	EPL	Protection concept	Group	T-class	Note
Pump motors P-201A, P-202A/B	Zone 2	Gc	Ex ec (increased safety, non-sparking; was Ex nA) or Ex db	IIA	T3	motor surface temperature and the start (t_E) time for Ex eb if Zone 1
Transmitters, RTDs, analyzers in the field	Zone 2 (Zone 1 at sample points)	Gb	Ex ia (intrinsically safe) via barriers / isolators	IIA	T4	IS loops need an entity / FISCO check
Solenoid valves of XV-201/202/209, positioners	Zone 2	Gb/Gc	Ex mb or Ex db (solenoids), Ex ia (positioners)	IIA	T4	
Junction boxes, cable glands	Zone 2	Gc	Ex eb boxes, Ex db / eb glands	IIA	T6	
Lighting fittings, local control stations	Zone 2	Gc	Ex ec / Ex db fittings; Ex db eb push-button stations	IIA	T3	LED fittings run cool
Analyzer house equipment	Zone 2 (inside)	Gc	Ex p (pressurised) cabinets or Ex ec; gas detection trips non-Ex power	IIA	T3	
Steam turbine of P-201B	Zone 2	Gc	mechanical equipment: ISO 80079-36/-37 (Ex h); governor electrics Ex db	IIA	T3	LP-steam surfaces ≤ 175 °C

6 · What fails on power loss

Consumer lost	Supply	Immediate effect	Designed response / mitigation
P-201A feed pump motor	MCC bus A	Feed to C-201 stops; V-201 stops emptying.	FSLL-201 starts the steam-turbine spare P-201B, which needs no electric power (diverse drive) — only its solenoid and governor, on UPS.
P-202A reflux pump motor	MCC bus A	Reflux and propane product stop; V-202 starts to fill.	PDSL-230 starts P-202B; that works only if P-202B is fed from another bus (EE-01). With both lost: the reflux-failure relief case.
DCS (UPS-DCS)	UPS, 30 min battery	Control outputs and screens lost when the battery runs down.	On loss of signal each control valve goes to its fail position (FC or FO, instrument list); the operator loses the view of the unit.
SIS (UPS-SIS)	UPS, 30 min battery	The SIS is de-energise-to-trip.	Loss of SIS power closes XV-201, XV-202 and XV-209: feed and heat off, the safe state.
Analyzer house AH-219/220	bus B	AI-219 / AI-220 freeze or fail.	No trip; product quality is followed by the tray temperatures until the analyzers return.
Lighting LTG-200	bus B + battery fittings	Normal lighting lost.	Battery fittings (90 min, assumed) light the escape routes and the local panels.
Site-wide power failure (outside the unit)	grid / site substation	Cooling-water pumps, instrument-air compressors and all unit motors stop together.	Cooling water lost to E-202 and every valve to its fail position at once — the general power failure relief case of the column (Relief & vent tab, EE-02).

7 · Checks against the model and the other documents

17 ✓, 0 ✗ (recomputed live in the Electrical tab of the simulator page).

Check	Load list	Model / other document	Result	Note
P-201A motor ≥ absorbed at the rated point × API 610 margin	75 kW	56.2 × 1.1 = 61.8 kW	✓	
P-202A motor ≥ absorbed at the rated point × API 610 margin	30 kW	22.0 × 1.25 = 27.5 kW	✓	
P-202B motor ≥ absorbed at the rated point × API 610 margin	30 kW	22.0 × 1.25 = 27.5 kW	✓	
P-201A motor ≥ absorbed at maximum feed	75 kW	54.1 kW	✓	
P-201A load factor at design 50–100 %	66 %	assumed efficient range	✓	
P-202A motor ≥ absorbed at maximum feed	30 kW	21.7 kW	✓	
P-202A load factor at design 50–100 %	69 %	assumed efficient range	✓	
P-201B (steam turbine) carries no electric load	0 kW	pump duty 49.7 kW	✓	diverse drive
Cable P-201A (150 m, 400 V)	50 mm²: Iz 138 A ≥ 129.5 A, ΔV 3.97 / 12.26 %	≤ 5 % running, ≤ 15 % starting	✓	
Cable P-202A (160 m, 400 V)	16 mm²: Iz 72 A ≥ 53.3 A, ΔV 4.86 / 12.40 %	≤ 5 % running, ≤ 15 % starting	✓	
Cable P-202B (170 m, 400 V)	25 mm²: Iz 91 A ≥ 53.3 A, ΔV 3.32 / 9.05 %	≤ 5 % running, ≤ 15 % starting	✓	
Specified T3 (200 °C) below the lowest AIT	372 °C	T3 = 200 °C; hottest allowed T2	✓	
Gas group IIA covers the fluids (no hydrogen)	IIA	IIA specified	✓	EE-03 if hydrogen is present
Hot surface below the AIT: LP steam design	175.0 °C	372 °C	✓	
Hot surface below the AIT: C-201 bottoms	109.0 °C	372 °C	✓	
UPS-DCS load ≤ 80 % of the rating	3.0 kW	5 kVA × PF 0.95	✓	
UPS-SIS load ≤ 80 % of the rating	1.0 kW	2 kVA × PF 0.95	✓	

8 · Assumptions

- Pump and motor efficiencies, power factors, demand factors, UPS efficiency and autonomy as section 1.
- Allowances: UPS-DCS 3.0 kW, UPS-SIS 1.0 kW, analyzer house 6 kW, lighting 10 W/m² (6 kW), small power 20 kW intermittent, motor heaters 0.3 kW; no heat tracing.
- Cable lengths P-201A 150 m, P-202A 160 m, P-202B 170 m; ambient 40 °C; 4 cables grouped; ampacity and resistance tables as typical published values.
- Hazardous-area extents typical values; auto-ignition temperatures typical published values (propane 450, n-butane 372, isobutane 460, methane 600, hydrogen 560 °C).
- Short-circuit withstand, protection coordination, earthing and lightning protection not covered.

9 · Engineering queries raised

No.	Subject	Query	Proposed answer	Status
EE-01	P-202A / P-202B power supply	Both reflux pumps are motor driven (30 kW each). If they share one MCC bus, a bus fault or a voltage dip stops both and the PDSL-230 auto-start cannot help.	Feed P-202A and P-202B from different bus sections (A / B) of the MCC with a bus-tie; consider automatic re-acceleration after short dips.	OPEN
EE-02	General power failure as a relief case	A site power failure stops the cooling-water pumps, the instrument-air compressors, P-201A and P-202A/B together: E-202 loses its cooling while every valve goes to its fail position (FV-204 and PV-210 open).	Relief & vent: confirm the total power failure case for PSV-201 and the flare header (usually it governs the overhead relief load).	OPEN
EE-03	Non-condensables and the gas group	The model's non-condensables are described as methane / hydrogen; at design they are 59 mol % of the vent gas leaving V-202. Methane keeps the area in group IIA; hydrogen above roughly 30 vol % in the released gas would move the vent and fuel-gas areas to IIB + H2 or IIC.	Process to give the composition of the non-condensables; until then specify IIB T3 for equipment around 1"-P-206, PV-210 and V-202.	OPEN
EE-04	P-201B turbine auto-start without power	The steam-turbine spare is the diverse drive for a power failure, but its trip-and-throttle solenoid, governor and the FSLL-201 logic need power.	Feed the solenoid and governor from UPS-DCS (done in this load list) and design the turbine start valve so that the auto-start works on UPS alone; test the start with the MCC dead during commissioning.	OPEN
EE-05	Cable routes, MCC location and fault level	Cable lengths (P-201A 150 m, P-202A 160 m, P-202B 170 m) are assumed: there is no plot plan with the substation. Short-circuit withstand and protection settings are not checked.	Electrical to confirm the substation location, the 400 V fault level and the protective devices, then re-run the cable sizing (thermal short-circuit check $S \geq I \cdot \sqrt{t} / k$).	OPEN